



Event → Check → Action

A space that reacts

이벤트가 생기면 확인하고 행동하는 패턴

BIG PICTURE

One pattern, everywhere



Player acts → code checks → something happens.

모든 반응형 코드의 기본 흐름

HOW TO COUNT

Count the axis — one block, one number



1 Stand on the start block — that spot is $(0, 0, 0)$.

2 $x \rightarrow$ walk **right**. Add **+1** for every block.

3 $y \rightarrow$ jump **up**. Add **+1** for every block.

4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

The three-step pattern

1 • EVENT

player walks somewhere

2 • CHECK

what block is there?

3 • ACTION

door opens / trap fires

이벤트 → 확인 → 행동

CONCEPT

Event + check together

```
def on_walk():  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("Welcome to my world!")  
player.on_travelled(on_walk)
```

Event fires every step — the **check** decides if the action runs.

이벤트는 매번, 행동은 조건이 맞을 때만

CONCEPT

Action can be a function

```
def spring_trap():  
    agent.place(AIR, DOWN)  
    player.say("You fell in my trap!")  
  
def on_walk():  
    if agent.inspect(DOWN) == RED_CARPET:  
        spring_trap()  
player.on_travelled(on_walk)
```

행동을 함수로 깔끔하게 정리해요

WATCH IT!

Step → check → react

```
below = RED_CARPET
```

```
below = BLUE_CARPET
```

```
inspect(DOWN) == RED_CARPET
```

```
✓ true → spring_trap()
```

```
✗ false → nothing – safe
```

▶ Result

확인 결과에 따라 행동이 달라져요

PREDICT 🤔

When does it fire?

Predict first — then click reveal

```
def on_walk():  
    if agent.inspect(DOWN) == GOLD_BLOCK:  
        player.say("Welcome!")  
player.on_travelled(on_walk)  
# player walks on grass, then gold
```

Reveal output 🔍 reveal

VOCAB

New words today

react respond to what the player does

check an **if** test before acting

action what runs when the check passes

pattern a repeatable code shape

반응하다

확인

행동

패턴



What's wrong with this code?

Goal: the door opens **only after** checking it's really there.

buggy

```
def on_walk():  
    open_door()  
    if agent.detect(DETECT_BLOCK, FORWARD):  
        player.say("Door found!")  
player.on_travelled(on_walk)
```

The check and action are swapped.



DEBUG

The bug

Bug: the action runs **before** the check. Check first, then act inside the `if`.

fixed

```
def on_walk():  
    if agent.detect(DETECT_BLOCK, FORWARD):  
        open_door()  
        player.say("Door found!")  
player.on_travelled(on_walk)
```

YOUR TURN!

Build a reacting space

✓ Use event → check → action once

✓ Pick a trigger block (carpet, gold...)

✓ Check the block before acting

✓ Make the action a function

✓ Send a video of your space reacting



QUIZ

QUESTION 1 OF 3

The door opens before the check runs. Fix?

Move the action inside the `if` — check first.

Open the door twice.

Rename the function.

Delete the check.



QUIZ

QUESTION 2 OF 3

A message appears on **every** step. What's missing?

More carpet.

Another event.

The **check** — an **if** that tests the block.

A bigger world.



QUIZ

QUESTION 3 OF 3

What does **event** → **check** → **action** mean?

Build, then delete, then rebuild.

Three chat commands in a row.

Walk, run, jump.

Something happens, code tests a condition, then it reacts.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Your world responds!

Next: auto-build a whole space.

다음 주: 명령 하나로 공간 전체 짓기